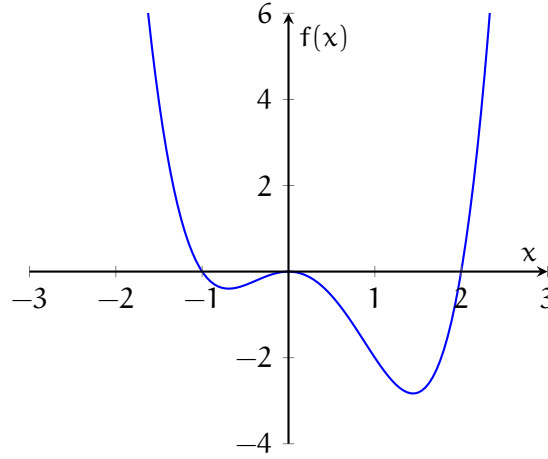


Yes. Any global minimum is also a local minimum. If the function is convex, a local minimum is also a global minimum. If not, there may be local minima that are not global. An example of such a function could be

$$f(x) = -2x^2 + x^4 - x^3.$$



We use the optimality conditions to identify the maxima and minima of the function. The first derivative of $f(x)$ is

$$f'(x) = -4x + 4x^3 - 3x^2.$$

If we solve $f'(x) = 0$, we obtain three solutions:

$$x_1 = 0, \quad x_2 = -0.693, \quad x_3 = 1.443.$$

These are the only candidates to be minima or maxima.

- Consider the interval $[-0.1, 0.1]$. x_1 reaches the maximum of f in this neighborhood. It is therefore a local maximum.
- Consider the interval $[-0.7, -0.6]$. x_2 reaches the minimum of f in this neighborhood. It is therefore a local minimum.
- Consider the interval $[1.4, 1.5]$. x_3 reaches the minimum of f in this neighborhood. It is therefore a local minimum.

This can also be verified using the second derivative of the function $f(x)$:

$$f''(x) = -4 + 12x^2 - 6x.$$

Indeed, if we substitute the solutions x in the function $f''(x)$ we obtain:

$$\begin{aligned} f''(x_1) &= -4 + 12(0)^2 - 6(0) = -4 &< 0 \Rightarrow \text{local maximum,} \\ f''(x_2) &= -4 + 12(-0.693)^2 - 6(-0.693) = 5.92 &> 0 \Rightarrow \text{local minimum,} \\ f''(x_3) &= -4 + 12(1.443)^2 - 6(1.443) = 12.32 &> 0 \Rightarrow \text{local minimum.} \end{aligned}$$

Since the function is a polynomial, we have

$$\begin{aligned} \lim_{x \rightarrow \infty} f(x) &= \lim_{x \rightarrow \infty} -2x^2 + x^4 - x^3 = +\infty, \\ \lim_{x \rightarrow -\infty} f(x) &= \lim_{x \rightarrow -\infty} -2x^2 + x^4 - x^3 = +\infty. \end{aligned}$$

Therefore, the function has no global maximum. Its global minimum is the local minimum associated with the lowest value of f . We can conclude that the function has

- a local maximum at $x_1 = 0$, $f(x_1) = 0$,
- a local minimum at $x_2 = -0.693$, $f(x_2) = -0.554$, and
- a global minimum at $x_3 = 1.443$, $f(x_3) = -2.77$.